BURNSIDE'S LEMMA AND APPLICATIONS IN CONTEST PROBLEMS

TAIKI AIBA

Abstract. In this paper, we will first examine and analyze a proof of Burnside's Lemma, then go over famous problems where the lemma is commonly used, and finally provide solutions to problems found in mathematical competitions where the lemma may be used (even if the lemma was not intended to be used from the problem author's perspective).

1. Introduction

Burnside's Lemma, also referred to as the Cauchy-Frobenius Theorem, is a lemma found in group theory that takes advantage of symmetry in groups to enumerate mathematical objects [6]. Although rooted in group theory, Burnside's Lemma is also a powerful enumerative combinatorial lemma that allows us to solve problems involving counting the number of ways to color the faces of a cube or arrange the beads of a necklace given additional restrictions on whether rotations and reflections give unique ways. In this paper, we will first examine and analyze a proof of Burnside's Lemma, then go over famous problems where the lemma is commonly used, and finally provide solutions to problems found in mathematical competitions where the lemma may be used (even if the lemma was not intended to be used from the problem author's perspective).

2. Preliminary Definitions

Before we introduce the statement and proof of Burnside's Lemma, we shall first go over a few definitions. Knowledge of the definition of a group and basic group theorems in abstract algebra is assumed, as well as notation commonly used in group theory [5].

Definition 2.1 (Group Actions)**.** A group G *acts* on a set X if there exists a map $\cdot : G \times X \to X$ such that $e \cdot x = x$ for all $x \in X$ and $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$.

Notably, this definition tells us that we can represent a map with a binary-operation-like notation that crosses a group with an arbitrary set. In each of the following definitions, it is assumed that a group G acts on a set X per the definition above [5].

Definition 2.2 (Fixed Point)**.** A *fixed point* of an element $g \in G$ is an element $x \in X$ such that $g \cdot x = x$.

Definition 2.3 (Stabilizer)**.** For any element $x \in X$, the *stabilizer* of x is denoted by G_x, where G_x is the set of all $g \in G$ such that $g \cdot x = x$.

Definition 2.4 (Orbit)**.** For any element $x \in X$, the *orbit* of x in X is denoted by $G \cdot x$, where $G \cdot x$ is the set of all $g \cdot x$ with $g \in G$.

3. Orbit-Stabilizer Theorem

The definitions concerning the orbit and stabilizer of an element of a set will be used to prove the following theorem, aptly named the Orbit-Stabilizer Theorem [9].

Theorem 3.1 (Orbit-Stabilizer Theorem)**.** *Let G be a finite group that acts on a set X. Let G_x be the stabilizer of an element $x \in X$, and suppose that the orbit $G \cdot x$ of x is finite. Then, the index $[G : G_x]$ is finite and equal to $|G \cdot x|$. This means that $|G_x| \cdot |G \cdot x| = |G|$ by Lagrange's Theorem.*

We will go over a proof of the Orbit-Stabilizer Theorem that introduces a function mapping the orbit to the left coset space of G_x. Showing that the function is bijective yields the result [9].

Proof. We will reuse the same notation as in the theorem statement. Let $x \in X$, and let G/G_x be the set of all left cosets of G_x. Define a function $\phi : G \cdot x \to G/G_x$ by $\phi(g \cdot x) = gG_x$. We will prove that ϕ is a bijection. To do so, we must show that ϕ is well-defined, injective, and surjective.

We must first show that ϕ is well-defined. Suppose that $g \cdot x = h \cdot x$ for some $g, h \in G$. Then,

$$\begin{aligned} h^{-1} \cdot g \cdot x &= h^{-1} \cdot h \cdot x \\ (h^{-1}g) \cdot x &= (h^{-1}h) \cdot x \\ (h^{-1}g) \cdot x &= e \cdot x \\ (h^{-1}g) \cdot x &= x. \end{aligned}$$

This shows that $h^{-1}g \in G_x$, which implies that $gG_x = hG_x$, so ϕ is well-defined.

To show injectivity, suppose that $f(g_1 \cdot x) = f(g_2 \cdot x)$ for distinct $g_1, g_2 \in G$. Then, $g_1G_x = g_2G_x$, which means that $g_2^{-1}g_1 \in G_x$. By definition, this means that $(g_2^{-1}g_1) \cdot x = x$. Furthermore,

$$\begin{aligned} g_2 \cdot (g_2^{-1}g_1) \cdot x &= g_2 \cdot x \\ (g_2g_2^{-1}g_1) \cdot x &= g_2 \cdot x \\ g_1 \cdot x &= g_2 \cdot x. \end{aligned}$$

This shows that ϕ is injective.

To show surjectivity, note that for every $gG_x \in G/G_x$, there exists $g \cdot x \in G \cdot x$ such that $\phi(g \cdot x) = gG_x$, so ϕ is surjective.

We have shown that ϕ is well-defined, injective, and surjective, so ϕ is a bijection. This implies that $|G \cdot x| = |G/G_x| = [G : G_x]$, and by Lagrange's Theorem, $|G_x| \cdot |G \cdot x| = |G|$. □

4. Burnside's Lemma

With preliminary definitions and the Orbit-Stabilizer Theorem in our hands, we are now ready to introduce Burnside's Lemma itself [2].

Lemma 4.1 (Burnside's Lemma)**.** *Let G be a finite group that acts on a set X. Let X/G be the set of all orbits of X. For any element $g \in G$, let X^g denote the set of all fixed points of g. Then,*

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|.$$

We will now present a proof of Burnside's Lemma that uses the Orbit-Stabilizer Theorem [2].

Proof. Consider the sum $\sum_{g\in G} |X^g|$. We may rewrite this in terms of the definition of X^g as follows:

$$\begin{aligned}\sum_{g\in G} |X^g| &= \sum_{g\in G} |\{x \in X : g \cdot x = x\}| \\ &= |\{(g,x) : g \in G, x \in X, g \cdot x = x\}| \\ &= \sum_{x\in X} |\{g \in G : g \cdot x = x\}| = \sum_{x\in X} |G_x|.\end{aligned}$$

By the Orbit-Stabilizer Theorem, $|G_x| = \frac{|G|}{|G\cdot x|}$. Thus,

$$\sum_{x\in X} |G_x| = \sum_{x\in X} \frac{|G|}{|G\cdot x|} = |G| \sum_{x\in X} \frac{1}{|G\cdot x|}.$$

It remains to rewrite this sum in our desired form.

Consider an orbit O, and consider all $x \in O$. Each x contributes $\frac{1}{|O|}$ to our sum, and there are $|O|$ such x. Thus, each orbit contributes $\frac{1}{|O|} \cdot |O| = 1$ to our sum, which boils our sum down into the number of orbits, which is $|X/G|$. Therefore,

$$\sum_{g\in G} |X^g| = |G| \sum_{x\in X} \frac{1}{|G\cdot x|} = |G| \cdot |X/G|,$$

from which rearranging gives us the desired form. □

As demonstrated, our presented proof of Burnside's Lemma makes powerful use of the Orbit-Stabilizer Theorem. Other than the initial rewriting of the sum $\sum_{g\in G} |X^g|$ in terms of $x \in X$, a key part of the proof that leads us to the final result is the consideration of the contribution of each x in each orbit O. This is an example of double counting, which is a common proof technique in combinatorics that involves counting the same sum in two different ways [4]. In this proof, double counting allows us to boil down a complicated sum into terms that we can use more easily.

5. Famous Applications of Burnside's Lemma

Now that we have presented and analyzed a proof of Burnside's Lemma, we will examine two problems where the lemma is famously used: counting colorings of the faces of a cube and counting arrangements of the beads of a necklace.

We will start by considering the problem where we color the faces of a cube. Suppose that we wanted to count the number of ways to color the faces of a cube with six different colors so that each face is painted exactly one unique color, and two colorings are considered identical if they can be obtained by rotating (but not reflecting) the cube [3].

Burnside's Lemma allows us to consider the problem from a group theory perspective. Let G be the group of all rotations of the cube. We can calculate $|G|$ by noting that there are six ways to choose the top face and four ways to choose the front face, for a total of $6 \cdot 4 = 24$ rotations. Let the set X being acted upon as the set of all colorings of the cube without rotation. Then, it is clear that $|X| = 6! = 720$.

If $g \in G$ were not the identity rotation, then there are no fixed points because any rotation of the cube will give a different coloring. If g were the identity rotation, then $X^g = X$ because every $x \in X$ is fixed. Thus, by Burnside's Lemma, there are $\frac{1}{24} \cdot 720 = \mathbf{30}$ colorings of the cube.

We will now consider another famous problem where Burnside's Lemma may be used: arranging the beads of a necklace. Suppose that we wanted to count the number of ways to arrange p beads on a necklace, where p is a prime number, such that each bead is either black or white, and two arrangements are considered identical if they can be obtained by rotating (but not reflecting) the necklace [1].

Let G be the subgroup of the symmetric group S_p generated by the cyclic permutation $\sigma = (1\ 2\ \cdots\ p)$. We see that $|G| = p$ because the order of this cyclic permutation is p by inspection. Let the set X being acted upon be the set of all arrangements of the necklace without rotation. For each of the p beads, there are 2 choices for its color, so $|X| = 2^p$.

We now count the number of fixed points. If $g = \sigma^k \in G$, where $k \in \mathbb{N}$ were not the identity permutation, then it is clear that the arrangements of all black and all white beads are fixed. The proof that every other arrangement is not fixed takes advantage of the fact that p is prime.

If there existed a fixed point $x \in X$ such that $g \cdot x = x$, then $g^n \cdot x = \sigma^{kn} x$ for any $n \in \mathbb{N}$. In particular, if $kn \equiv r \pmod p$, for $0 \leq r < p$, then the permutation $g = (a_1\ a_{1+r}\ a_{1+2r}\ \cdots\ a_{1+pr})$ (indices taken modulo p) would need to map to the same permutation as the original for all possible residues r modulo p. Since p is prime, no matter what k is, the set of residues $\{1, 1+r, 1+2r, \ldots, 1+pr\}$ will form a complete residue class modulo p if $r \neq 0$, so for example a_1 can be in any position of the permutation by varying r. If $r = 0$, then we have the identity again. This means that we require every bead in x to be the same color; i.e., they are all black or all white. Thus, there are $p-1$ permutations in G that are not the identity, and each one has two fixed points.

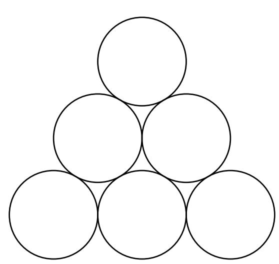
If g were the identity permutation, then there are 2^p fixed points as every arrangement in X is fixed. Thus, by Burnside's Lemma, there are $\frac{1}{p} \cdot (2^p + 2(p-1)) = \frac{2^p+2p-2}{p}$ arrangements.

As suggested above, it is an important fact in this solution that p is prime. Note that if p were not prime, then the number of fixed points cannot be calculated in this way. Indeed, if $p = 9$, then consider the permutation $(a_1\ a_4\ a_7)(a_2\ a_5\ a_8)(a_3\ a_6\ a_9)$. If we have an arrangement of 9 beads where there is a pattern of 3, e.g., black, black, white, black, black, white, black, black, white, then we can see that this is a fixed point of the permutation.

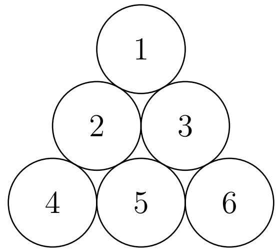
6. Applying Burnside's Lemma in Competition Problems

Although not a significantly-used lemma in most middle and high school competitions, Burnside's Lemma can be used as valid solutions to solve a number of counting problems. We will go over two problems from math competitions where we may use Burnside's Lemma. We will start with a fairly simple problem to demonstrate [8].

Problem (2017 AMC 12B, Problem 13)**.** In the figure below, 3 of the 6 disks are to be painted blue, 2 are to be painted red, and 1 is to be painted green. Two paintings that can be obtained from one another by a rotation or a reflection of the entire figure are considered the same. How many different paintings are possible?



Solution. Number the disks from 1 to 6 going from top to bottom and from left to right.



Let G be the set of all rotations and reflections of the diagram. There are 6 of these in total: 3 rotations – $0°$, $120°$, and $240°$ – and 3 different reflections, which are over the lines passing through the centers of (1) and (5), (2) and (6), and (3) and (4). Thus, $|G| = 6$.

Let the set X being acted upon be the set of all diagrams. The $0°$ rotation has the total number of diagrams fixed, which is $\binom{6}{3} \cdot \binom{3}{2} = 60$. The $120°$ and $240°$ rotations each have zero fixed diagrams, as one would need the same color for disks (1), (4), and (6) and the same color for disks (2), (3), and (5), which is not possible.

Consider the reflection over the line passing through disks (1) and (5). The green can be placed in 2 ways – disk (1) or disk (5) – and then the blues must be placed in either disks (2) and (3) or disks (4) and (6), so the number of fixed diagrams is $2 \cdot 2 = 4$. Similarly, the other two reflections also have 4 unique fixed diagrams. Thus, by Burnside's Lemma, we get $\frac{1}{6} \cdot (60 + 4 + 4 + 4) = \mathbf{12}$ different paintings.

We will now solve an interesting algebraic counting problem with an unexpected use of Burnside's Lemma [7].

Problem (2010 AIME II, Problem 10)**.** Find the number of second-degree polynomials $f(x)$ with integer coefficients and integer zeros for which $f(0) = 2010$.

Solution. Consider a second-degree polynomial $f(x)$. If r and s are the zeros of the polynomial and a is the leading coefficient, then we may write

$$f(x) = a(x - r)(x - s).$$

Since r and s are indistinguishable, we may define the permutation group G that consists of the identity permutation and the permutation that transposes r and s. We may then define the set X being acted upon as the set of integer triples (a, r, s) such that $f(0) = ars = 2010$.

There are $4 \cdot 3^4$ fixed points from the identity permutation by distributing the prime factors of $2010 = 2 \cdot 3 \cdot 5 \cdot 67$ among a, r, and s in 3^4 ways and four ways to assign their signs since we can have a, r, and s be all positive, a and r be negative, r and s be negative, or a and s be negative. We also have 2 fixed points from the transposition permutation, namely $r = s = 1$ and $r = s = -1$. This is because we want $r = s$, which means that the product rs must be a perfect square. The only perfect square divisor of 2010 is 1, which yields the two fixed points. Thus, by Burnside's Lemma, there are $\frac{1}{2} \cdot (4 \cdot 3^4 + 2) = \mathbf{163}$ ordered triples (a, r, s).

7. Concluding Remarks

Burnside's Lemma is a beautiful result that combines concepts covered in group theory with enumerative combinatorics. When we want to count orderings and we wish to avoid annoying casework or constructive counting, we may often wish to resort to using Burnside's Lemma. From solving simple problems like coloring the faces of a cube, to counting polynomials with integer coefficients and roots with a specific constant term, Burnside's Lemma may appear in our lives in ways we do not even expect. We just need a group and a set, and we are all set.

References

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